

Real Time Traffic Light Timing Optimisation Using Reinforcement Learning

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Abstract—A major problem especially in urban areas is traffic congestion which causes increased travel times, pollution, and compromises safety. This paper presents the design of the Smart Traffic light timing Management System, optimized in real-time traffic light timings using reinforcement learning techniques. This system scans data from different sources, including vehicle count in a network and waiting time of the vehicle at a traffic stop, which are utilized to continuously to alter traffic signals based on current situations. Utilizing reinforcement learning algorithms, the system continuously learns and adapts to changing traffic scenarios, striving for minimal congestion. The simulations run in SUMO (Simulation of Urban Mobility) indicate a noticeable reduction in wait times and improved flow across roads. The results form the basis for the inclusion of advanced data analytics in traffic management: it represents an effective method of mitigating urban congestion problems. The study concludes that by leveraging real-time data and machine learning, the traditional traffic systems can be transformed into an intelligent responsive network, capable of enhancing the efficiency and safety of urban transportation.

Keywords—Urban Traffic Management, Reinforcement Learning, Smart Signal Timing, Machine Learning, Traffic Congestion Control

I. INTRODUCTION

Rapid urbanization of cities has marked the exponential rise in traffic congestion with further consequences that include heavy delays, fuel wastage, environmental pollution, and even increased accident possibilities. In addition, traditional traffic management systems, mostly operating on static signal timing, cannot cope up with changing dynamics of the traffic and variable densities at different intersections. Furthermore, these systems often worsen congestion, especially during peak time.

There is an emerging need to provide intelligent solutions for managing traffic to respond to changing real-time traffic conditions and optimize the flow of traffic. Developments in machine learning and data analytics have set the stage for the evolution of smart traffic systems that can make decisions in real time based on the collection of traffic data provided by numerous sensors and cameras.

A Smart Traffic Management System- dynamic adjustment of traffic signal timings based on real-time traffic data using machine learning is presented in this paper. In this, with reinforcement learning, the system learns optimal signal timings for minimal waiting times for vehicles,

congestion reduction, and overall optimization in traffic movement. The system thus ultimately aims to improve road safety by reducing congestions and emission and then adjusting the traffic signals accordingly. This approach provides flexible, scalable solutions to the complex problem of urban traffic management in an efficient manner.

II. LITERATURE REVIEW

Hazarika (2024) proposed edge machine learning technique for smart traffic management in intelligent transportation systems. This system controls the real-time traffic signals based on traffic density with provisions of giving priority levels to emergency vehicles using the YOLO. Our project objective is optimization in real-time from IoT sensors that are targeting congestion relief and enhancement in the flow of traffic[1].

Zhang introduced the integrated framework for real-time intelligent traffic management on smart highways. The authors used deep learning, namely, LSTMs, for forecasting the volume of traffic and optimizing flow using variable speed limits and ramp metering. For this specific approach narrowed down to highways, the best suited for our purpose of real-time data-driven dynamic adjustment of traffic signals in urban congestion management[2].

System Review of Different Nigam, 2023 reviewed the different aspects of Intelligent Traffic Management Systems (ITMS), with specific mentions including computer vision and AI as facilitating elements in the control of traffic. Like our project, this review highlights the role of machine learning and actual data acquired real-time in managing congestion so as to control urban traffic better through imaging and vehicle analysis[3].

Ali (2023) designed a real-time dynamic, smart traffic light system focusing on efficiency in reducing congestion. This research work, as it seems, is very much related to our approach towards the project, because optimization of traffic signals for minimal waits and maximum flow efficiency is carried out using vehicle counts[4].

Alsheikhy, 2023: Developed an intelligent adaptive algorithm for smart traffic management using real-time traffic data from IoT sensors and a CNN-based deep learning technique. The system aims to dynamically optimize traffic flow in reducing congestion, enhancing road safety, just like our project aims to dynamically optimize traffic flow, reduce congestion, and enhance road safety[5].

Musaproposed (2023) an IoT-and machine learning-based sustainable traffic management system for smart cities by promoting improved real-time traffic flow and enhanced reduction of environmental impacts. This paper, just like our project, relies on sustainability as it speaks to effective systems meant to conserve fuel and reduce emission[6].

Sultana (2024) designed a smart traffic management system to improve mobility and timely responses towards emergencies. Their work, applying IoT and AI for real-time traffic control during peak hours, aligns with the focus areas of our project: decreasing congestion and improving traffic response times through adaptive signal management[7].

Rekha (2023) Designs a dynamic traffic management system, that can be integrated with YOLO for density detection from images. In this, the traffic system varies dynamically just like our use of machine learning and current traffic status that can avoid congestions and make smooth flows of traffic at urban junctions[8].

Johny (2023) developed machine learning integration for dynamic traffic management of smart cities. Their target to decrease congestion in cities as well as the optimization of patterns of traffic resemble our project, which predicts the pattern of traffic based on real-time data. Signal timings are also modified according to that[9].

Smith showed methods of AI-based optimization in smart cities for traffic, and it demonstrated the way to manage traffic signal in a dynamical way, in real time. That is exactly our system, and therefore, it focused on the improvement of urban traffic management through the use of intelligent algorithms that will lead to reduced delays and general efficiency in traffic[10].

III. EXISTING METHODS

Currently, traffic management systems are more sophisticated, making use of the latest technologies to ensure maximum efficiency with reduced congestion. What presently exists can be broadly classified under several approaches with varied techniques adopted for traffic control and optimization.

A. Traditionally Used Systems of Traffic Signal Control

Traditionally, the systems of traffic signal control have been either based on fixed-time or actuated control. Fixed-time systems operate in cycles that have been pre-set up for optimizing traffic, generally according to historical evidence. Even though they are simple, they cannot respond to real-time conditions hence implying some delay and increase in congestion levels. Actuated control systems, however, adjust their signal timing based on real time vehicle detection hence a more responsive approach. However, they still lack the predictive capabilities that would serve towards management of complicated urban traffic efficiently.

B. Adaptive Traffic Signal Control

As a proven alternative to overcome the limitations of the traditional solutions, adaptive traffic signal control has developed. These systems use real-time information from cameras, inductive loops, and sensors of the present traffic situation to change signal timings based on the current traffic conditions. For example, SCOOT (Split Cycle Offset Optimization Technique) and ATCS (Adaptive Traffic Signal Control System) are examples. These systems

enhance the movement of traffic and minimize delays by readjusting signal timings due to changing traffic patterns.

C. Intelligent Transportation Systems

An Integration of Intelligent Transportation Systems. The smart integration of advanced technologies like IoT, machine learning, and big data analytics are used to improve traffic management. In addition, data from multiple sources - GPS devices, mobile applications, social media, and more are utilized and leveraged by ITS solutions to deliver real-time information on traffic, optimizes routes, and helps predict trends in matters like traffic. The ultimate benefits of ITS are the increase efficiency, improvement of safety conditions, and enhancement of the environment.

IV. METHODOLOGY

In this work, an adaptive traffic management system that employs a deep Q-network, optimizing traffic light control in a simulated urban environment, is proposed. In this regard, the system has been designed with the support of the SUMO tool and the TraCI tool for simulating realistic conditions based on real-time vehicle flow data within an urban environment.

A. System Overview

The system is based on a deep Q-learning framework and dynamically controls traffic lights at the crossroads of several intersections. The traffic network was modelled in SUMO, and the state of the system is represented by the number of vehicles on different lanes approaching an intersection, while the available actions are configurations for different phases for traffic lights, such as moving the green phase from one lane to another.

B. Traffic Data Collection

The SUMO environment integrates with TraCI, which makes it possible to collect real-time traffic data. Vehicle counts are collected from every junction's controlled lanes by the application of `traci.lane.getLastStepVehicleIDs()` to calculate the number of vehicles toward every lane. Waiting times of vehicles are also retrieved using the function `traci.lane.getWaitingTime()`, which gives an indication of the traffic congestion level. All this information is used to update the state space for the reinforcement learning agent.

C. Reinforcement Learning Framework

1) Deep Q-Network

The reinforcement learning agent is introduced as a DQN. The agent is trained to minimize the waiting time at each intersection by learning the adjustment in the length and sequence of its phases of the traffic light. The neural network contains three fully connected layers with ReLU activation functions for introducing nonlinearity. In this scenario, the input layer will take state information from the vector that encodes information about the number of vehicles on each lane. The output layer then comprises Q-values for all possible actions, which the agent will use to make the best choice according to an epsilon-greedy strategy. Adam Optimizer is user here and we are implementing MSE loss to calculate the error between predicted Q-value and the target Q-value. The agent trains over epochs and adjusts its Q-values according to the rewards obtained at every step, which is inversely proportional to the waiting time of the vehicle at the junction.

2) Learning and Memory Management

For effective learning, the agent maintains a memory buffer for every junction. Transitions that it stores in this buffer include the current state, next state, action taken, reward received, and whether the episode has terminated. The transitions are sampled in batches, thus improving the efficiency of learning by removing temporal correlations between consecutive samples. Using an epsilon decay strategy, an agent balances the process of exploitation and exploration by adapting what would be an appropriate exploration strategy for it at a given time. It means at the beginning stages, it explores all sorts of actions with a high probability but reduces exploration over time as it develops confidence in what action to select.

D. Adaptive Traffic Control

Every junction's traffic light control is managed dynamically, but in phase of the traffic light by adjusting the one that has been allowed to change. This is dependent on the action chosen by the agent and gives an assessment based on the state of the traffic at each junction. Its determination is whether to extend the current phase or to shift over to another. The control of the traffic light in the city is applied through the function `traci.trafficlight.setPhaseDuration()`, sets the phase duration based on the decision of the agent.

E. Performance Evaluation

The proposed system is evaluated in terms of total waiting time of all vehicles at all junctions. The performance is measured in terms of reduced waiting time of the vehicle and improved efficiency in traffic flow across multiple simulation steps. Furthermore, the cumulative reward during every epoch of training is monitored to visualize the progression of learning. Plot of the cumulative waiting time versus epochs is enough to give the performance chart of the agent in terms of learning while the traffic management improves as shown in Figure 1.

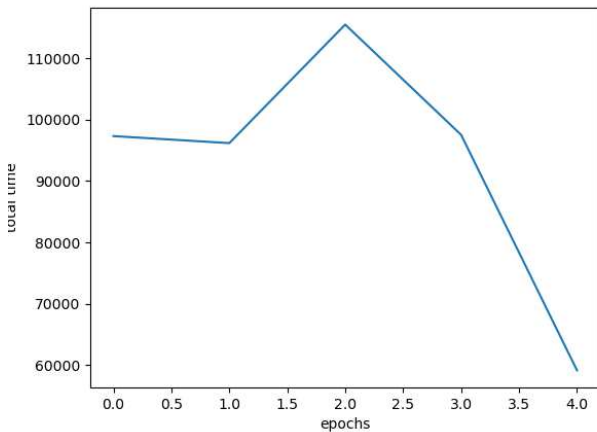


Fig. 1. Total Time versus Epochs

V. RESULTS

A. Experimental Configuration

For running the experimentation, two modes of SUMO simulations are used: one non-graphical mode for training the agent and the other is graphical mode for testing purposes with learned traffic control policies visualized. The epoch for

training is over 50, and simulation steps at 500 per epoch are taken. A comparison is done between a proposed adaptive system and the fixed-time traffic control system in order to show that this approach establishes well in helping to reduce congestion.

B. Results and Discussion

Percentage improvements are shown below to compare proposed RL model with the fixed-time traffic signal control system. Results are represented in Figure 2.

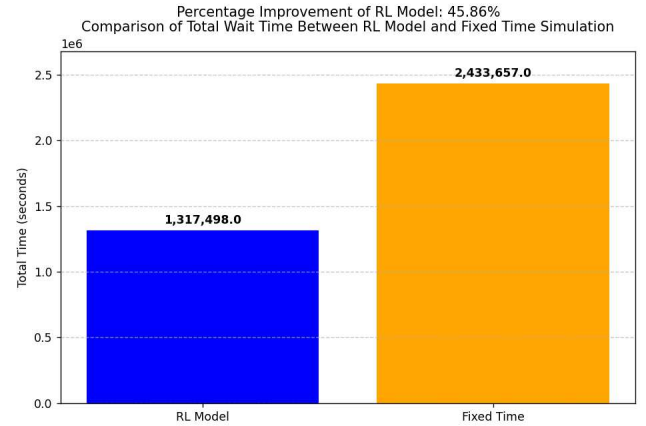


Fig. 2. Total Time comparison between fixed and RL Model

C. Percentage Improvement

The RL model had an overall traffic management performance that was better by 45.86% compared to the fixed-time system. This is attributed to the dynamic adjustment capability of the RL model in establishing traffic light timings under real-time traffic scenarios, which is quite far from a fixed-time system, whose setup is not dependent upon the volume of traffic and is changed only in pre-programmed time intervals.

D. Model Performance Discussion

The high improvement in performance by the RL model means that traffic flow can now be efficiently run without waiting times at intersections for vehicles. The RL-based system makes adaptations towards the variances of the traffic patterns, keeping its flow smooth while making overall travel times shorter. A fixed time system could not achieve that since it cannot take into account real-time traffic density; instead, it leads to longer queues and increased congestion.

This nearly 46% improvement percentage manifests the RL model's achievement in balancing the flow of traffic more effectively, which indeed gives the solution a better cutting-edge fit for modern urban environments where conditions are dynamic with time changes.

VI. CONCLUSION AND IMPLICATIONS

The results indicate that RL-based traffic control systems are a feasible alternative to fixed time systems, mainly for real-time, dynamic environments in cities. What the RL models can do for enhancing traffic management with reduced congestion and improved commuter experience is to ingest real-time data and update their models in real time based on the patterns of the traffic.

The results suggest that the possibility of smarter and adaptive control systems by coupling RL-based models with smart city infrastructures- IoT sensors, cameras, and V2I communication is there. Thus, travel times will be reduced along with some emission reduction and enhancement in road safety.

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